

Advanced techniques in applied economics

Lecture 4: Causal discovery and non-Gaussian identification

Piotr Zwiernik



**Universitat
Pompeu Fabra**
Barcelona

Statistics, Probability
and Machine Learning
Research Group



Barcelona School of Economics

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What can we learn from observational data alone?

What changes today?

In Lecture 3 we gave causal meaning to DAGs using factorization, d-separation, and interventions. Today we ask a harder question:

How much of the causal direction can be recovered without intervening?

Constraint-based discovery

Use conditional independence tests to recover the skeleton and the v-structures.

Extra identifying structure

Use assumptions such as non-Gaussianity or invariant mechanisms to go beyond CPDAGs.

Main question: Which arrow directions are visible from conditional independence alone, and which require extra assumptions?

Roadmap for today

1. constraint-based discovery and the PC algorithm
2. why Markov equivalence is the main obstacle
3. why linear Gaussian SEMs do not identify direction
4. LiNGAM and the role of non-Gaussianity
5. more realistic shocks: common volatility and mean independence
6. a brief link to additive noise models

Conditional independence narrows down the graph; asymmetry assumptions identify direction.

Part 1: Constraint-based discovery

Why conditional independence alone stops at a CPDAG

Constraint-based discovery uses the logic of Lecture 3:

- d-separation implies conditional independence,
- missing edges correspond to some separating set,
- colliders can be oriented from separating sets.

From observational data we may learn

- the **skeleton**,
- the **v-structures**,
- and therefore a **CPDAG**.

What this does not buy us

Without extra assumptions, we usually cannot orient every edge.

High-level idea of the PC algorithm

The PC algorithm is the canonical constraint-based discovery method.

Skeleton phase

1. Start from the complete undirected graph.
2. Test marginal and conditional independences.
3. Remove an edge whenever some separating set is found.

Orientation phase

1. Use separating sets to orient v-structures.
2. Apply graphical rules to orient additional edges.

Output

A **CPDAG**, not necessarily a unique DAG.

PC is a discovery procedure based on conditional independence, not on interventions.

For the general logic of constraint-based discovery, see Spirtes, Glymour, and Scheines, *Causation, Prediction, and Search*, Chapters 4–5.

The PC algorithm: skeleton phase

Input

Data on $X_1, \dots, X_m$, a conditional independence test, and a significance level α .

Skeleton phase

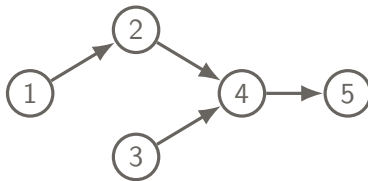
1. Initialize with the complete undirected graph on $\{1, \dots, m\}$.
2. For conditioning sets of size $0, 1, 2, \dots$:
 - ▶ for each adjacent pair (i, j) ,
 - ▶ test $X_i \perp\!\!\!\perp X_j \mid X_S$ for candidate sets S ,
 - ▶ if independence is not rejected, remove edge $i - j$ and store S .

Interpretation

Search for a small set of variables that explains away the apparent association between X_i and X_j .

A toy discovery example

Suppose the true DAG is



Skeleton phase intuition

- marginal tests may already remove $1 - 3$ and $2 - 3$,
- then conditioning on 2 may remove $1 - 4$ and $1 - 5$,
- conditioning on 4 may remove $2 - 5$ and $3 - 5$.

The skeleton is found by asking which pairs can be separated by some conditioning set.

Orientation phase: where colliders come from

Suppose the skeleton contains



with no edge between i and k . If the separating set for (i, k) *does not* contain node j , then PC orients

$$i \rightarrow j \leftarrow k.$$

Why?

If j were a non-collider, then j would have to belong to every minimal separator between i and k .

After orienting all such colliders, PC applies **Meek's rules** to propagate additional arrow directions while avoiding new v-structures and directed cycles.

For the orientation rules, see Meek (1995), *UAI*.

A first PC example in R

```
library(pcalg)
library(MASS)
set.seed(1)

Sigma <- matrix(c(1,.7,0,.49,0,
                  .7,1,0,.7,0,
                  0,0,1,.6,0,
                  .49,.7,.6,1,.7,
                  0,0,0,.7,1), 5, 5)

X <- mvrnorm(500, mu = rep(0, 5),
            Sigma = Sigma)
```

```
suffStat <- list(C = cor(X), n = nrow(X))
pc.fit <- pc(suffStat,
            indepTest = gaussCitest,
            alpha = 0.01,
            labels = paste0("X", 1:5),
            method = "stable")

pc.fit
```

Interpretation

For Gaussian data, the PC algorithm uses partial-correlation tests to recover a CPDAG.

For the `pcalg` implementation, see Kalisch et al. (2012), *Journal of Statistical Software*.

Why we should be cautious with PC

Constraint-based discovery is elegant, but it rests on demanding assumptions.

Statistical issues

- many CI tests,
- finite-sample error propagation,
- high-dimensional conditioning sets,
- sensitivity to near-unfaithfulness.

Economic issues

- latent confounding is common,
- Gaussianity may be implausible,
- observational data may identify only part of the structure.

PC is best viewed as a disciplined way to learn a CPDAG, not as a button that magically discovers the true causal graph.

Whenever possible, incorporate expert knowledge into the learning algorithm.

Part 2: Non-Gaussian identification

Why linear Gaussian SEMs do not identify direction

Take two centered Gaussian variables (X, Y) .

Forward representation

$$Y = aX + \varepsilon_Y, \quad \varepsilon_Y \perp\!\!\!\perp X.$$

Backward representation

$$X = bY + \varepsilon_X, \quad \varepsilon_X \perp\!\!\!\perp Y.$$

Key point

The same bivariate Gaussian distribution admits both directions as valid linear SEMs with independent noise.

Gaussianity is convenient for estimation, but it hides causal direction.

What kind of extra assumptions can break the symmetry?

To identify direction from observational data, we need an asymmetry that is invisible in the Gaussian model.

Three common strategies

- **Constraint-based methods:** rely on conditional independence and stop at a CPDAG.
- **Non-Gaussian linear models:** LiNGAM uses non-Gaussianity of the shocks.
- **Nonlinear noise models:** additive noise models use functional asymmetry.

Today's focus

We stay in the linear SEM world and ask what non-Gaussian shocks can reveal.

Part 3: LiNGAM and non-Gaussianity

LiNGAM assumptions

LiNGAM stands for **Linear Non-Gaussian Acyclic Model**.

Model

$$X = BX + \varepsilon,$$

where

- B is compatible with a DAG,
- the errors $\varepsilon_1, \dots, \varepsilon_m$ are mutually independent,
- the errors are non-Gaussian (except possibly one),
- there are no hidden common causes.

LiNGAM keeps linearity and acyclicity, but replaces Gaussian noise by non-Gaussian noise.

For the original LiNGAM result, see Shimizu et al. (2006), *Journal of Machine Learning Research*.

Why non-Gaussianity helps: $X \rightarrow Y$

In the Gaussian case, both directions may admit a linear model with independent noise.
With non-Gaussian noise, this symmetry breaks.

Bivariate intuition

Suppose

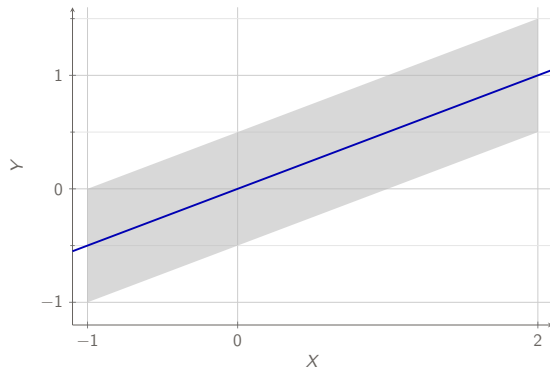
$$Y = aX + \varepsilon, \quad \varepsilon \perp\!\!\!\perp X,$$

with non-Gaussian X and ε . Then, in general, there is **no** backward linear model

$$X = bY + \tilde{\varepsilon}, \quad \tilde{\varepsilon} \perp\!\!\!\perp Y.$$

For LiNGAM identifiability, see Shimizu et al. (2006), *A Linear Non-Gaussian Acyclic Model for Causal Discovery*.

Why non-Gaussianity helps: forward model

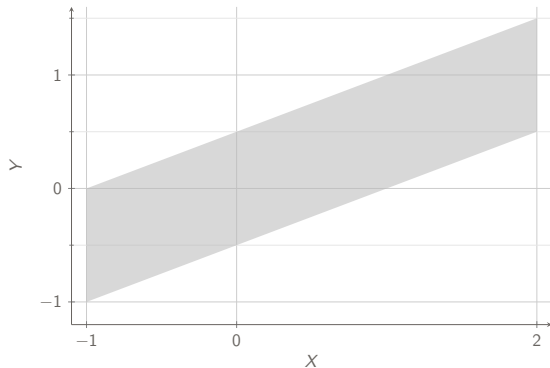


Consider the forward model

$$Y = 0.5X + \varepsilon_Y, \quad \varepsilon_Y \perp\!\!\!\perp X.$$

The gray strip is exactly the support implied by $\varepsilon_Y \sim U[-\frac{1}{2}, \frac{1}{2}]$ and $X \sim U[-1, 2]$.

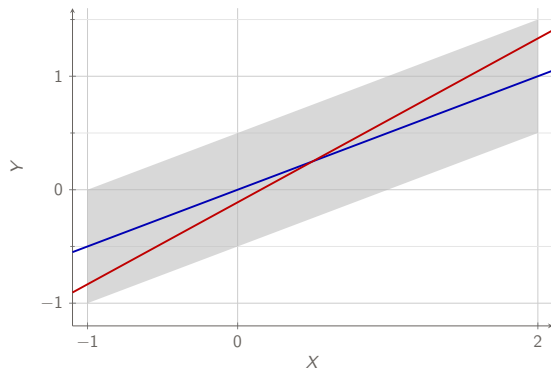
Why non-Gaussianity helps: reverse question



Question: can the same joint distribution arise from a backward linear model

$$X = bY + c + \varepsilon_X, \quad \varepsilon_X \perp\!\!\!\perp Y ?$$

Why non-Gaussianity helps: failed reverse fit



Let the red line be the least-squares fit for predicting X from Y :

$$(b, c) = \arg \min_{b, c} \mathbb{E}(X - bY - c)^2.$$

Even with the best linear backward fit, the residuals cannot be independent of Y .

DirectLiNGAM: the main idea

DirectLiNGAM looks for an exogenous variable first.

Key insight. If X_j has no parents, then $X_j = \varepsilon_j$. Now take another variable X_k . In a linear SEM it can be decomposed as

$$X_k = \beta_{kj}X_j + r_k^{(j)},$$

where the remainder $r_k^{(j)}$ depends only on the other structural shocks.

Thus, $X_j = \varepsilon_j$ is independent of $r_k^{(j)}$.

1. Find a variable whose regression residuals are as independent from it as possible.
2. Treat it as exogenous and remove its effect from the remaining variables.
3. Repeat on the residual system.

For DirectLiNGAM, see Shimizu et al. (2011), *Journal of Machine Learning Research*.

A simple LiNGAM simulation in R

```
set.seed(1)
n <- 2000

x1 <- rt(n, df = 5)
x2 <- 0.8 * x1 + rt(n, df = 5)
x3 <- -0.6 * x1 + 0.7 * x2 +
      rt(n, df = 5)

X <- cbind(x1, x2, x3)
colnames(X) <- c("x1", "x2", "x3")
```

```
library(pcalg)
fit <- lingam(X)
fit$Bpruned
```

Interpretation

The estimated matrix Bpruned gives the directed linear effects. Because the shocks are non-Gaussian, the causal ordering can become identifiable even though the model is linear.

Examples

- asset-return shocks are heavy-tailed,
- firm-level outcomes are skewed,
- macro shocks often have outliers and asymmetries.

Non-Gaussianity is not only a robustness issue; it can also be an **identification resource**.

Part 4: Beyond strict independence of shocks

Why independent shocks may be too strong in economics

The basic LiNGAM assumption is that the structural shocks are mutually independent. This can be too strong in economic applications.

A simple example

Consider the linear SEM

$$X_1 = \varepsilon_1, \quad X_2 = aX_1 + \varepsilon_2,$$

but suppose the shocks have the form

$$\varepsilon_1 = \sigma U_1, \quad \varepsilon_2 = \sigma U_2,$$

where U_1, U_2 are independent, non-Gaussian; and $\sigma \geq 0$ is a **random** common component.

What happens?

Conditional on σ , the shocks are independent. Marginally, however, they are dependent.

Economic interpretation This is a crude model of common market volatility:

- shocks need not share a mean factor,
- but their size may co-move.

Other reasons independence may fail

- sectoral exposure to the same aggregate news,
- omitted low-frequency macro conditions,
- equilibrium or strategic dependence across units.

A weaker asymmetry than full independence

In economics, full mutual independence of shocks is often too restrictive. A more realistic goal is to look for weaker asymmetry conditions that still distinguish cause from effect.

One candidate: mean independence

Instead of requiring full independence, one may ask whether the residual in the correct direction has conditional mean zero, while the reverse direction generically does not.

This is a research direction rather than a standard off-the-shelf method. The lesson for us is that identifiability may survive under weaker assumptions than classical LiNGAM uses.

A toy simulation with common volatility

```
set.seed(1)
n <- 2000

sigma <- abs(rnorm(n)) + 0.3
u1 <- rt(n, df = 5)
u2 <- rt(n, df = 5)

e1 <- sigma * u1
e2 <- sigma * u2
x1 <- e1
x2 <- 0.8 * x1 + e2
```

```
par(mfrow = c(1,2))
plot(x1, x2, pch = 16, cex = .4)
plot(abs(e1), abs(e2), pch = 16,
      cex = .4)
```

Interpretation

The second plot reveals common volatility in the shocks. This violates strict independence, even though the directional SEM remains meaningful.

Part 5: A brief look beyond linearity

Alternative route: Additive noise models

A different route to identifiability is nonlinear asymmetry.

Additive noise model

$$Y = f(X) + \varepsilon, \quad \varepsilon \perp\!\!\!\perp X.$$

Under suitable regularity conditions, the reverse representation

$$X = g(Y) + \tilde{\varepsilon}, \quad \tilde{\varepsilon} \perp\!\!\!\perp Y$$

usually does not exist.

Comparison with LiNGAM

LiNGAM uses linearity plus non-Gaussianity. Additive noise models use nonlinearity plus independent noise.

For additive noise models, see Hoyer et al. (2009), *NIPS*, and Peters, Janzing, and Schölkopf, *Elements of Causal Inference*, Chapter 5.

Extension: graphical ideas for time series

From static SEMs to dynamic graphical models

For a multivariate time series, a natural model is

$$X_t = A_1 X_{t-1} + \cdots + A_p X_{t-p} + \varepsilon_t.$$

Directed lagged structure: Zeros in the matrices A_ℓ mean that some lagged variables do not directly predict others. This is the dynamic analogue of missing directed edges.

Contemporaneous structure: If $\varepsilon_t \sim N(0, \Sigma_\varepsilon)$, then zeros in $K_\varepsilon = \Sigma_\varepsilon^{-1}$ encode conditional independence among shocks at time t .

Interpretation

A graphical time-series model separates

- **lagged directional effects** from
- **instantaneous conditional dependence of shocks.**

Why economists may care about graphical time series

These ideas are natural in macroeconomics and finance.

Lagged graph

A sparse lagged graph can summarize who predicts whom:

- sectoral spillovers,
- macro transmission channels,
- Granger-causal relations in large VARs.

Shock graph A sparse graph for ε_t can summarize contemporaneous residual links:

- common financial stress,
- instantaneous market co-movement,
- residual dependence after lagged dynamics are removed.

Dynamic graphical models ask: who predicts whom over time, and which shocks still interact contemporaneously?

For sparse VARs and graphical time-series ideas, see Basu and Michailidis (2015), *Annals of Statistics*, and the BigVAR package paper by Nicholson et al. (2020), *Journal of Statistical Software*.

A simple workflow in R

1. estimate a sparse VAR for lagged structure,
2. compute fitted values and residuals,
3. fit a Gaussian graph to the residuals.

```
library(BigVAR)
# X is a T x m matrix of series
mod <- constructModel(X, p = 1,
                      struct = "Basic",
                      gran = c(10, 10),
                      verbose = FALSE)

fit <- cv.BigVAR(mod)
B <- as.matrix(coef(fit))
Z <- as.matrix(fit@lagmatrix)
Yhat <- t(B %*% Z)
```

```
Yobs <- tail(fit@Data, nrow(Yhat))
Ehat <- Yobs - Yhat
S <- cov(Ehat)

library(glasso)
out <- glasso(S, rho = 0.05)

library(qgraph)
qgraph(-cov2cor(out$wi),
       layout = "spring")
```

Interpretation

The sparse VAR captures lagged directed effects. The residual precision matrix then estimates contemporaneous conditional dependence among the shocks.

Take-away

- Constraint-based discovery uses conditional independence to recover a CPDAG.
- Markov equivalence limits what can be learned from conditional independence alone.
- Linear Gaussian SEMs do not identify direction in general.
- LiNGAM uses non-Gaussianity to restore identifiability.
- For economic data, strict independence of shocks may be unrealistic; weaker asymmetries such as mean independence can be more credible.
- Beyond linearity, additive noise models provide another route to directional identification.

Conditional independence narrows down the graph. Asymmetry assumptions identify direction.

Looking ahead

Next lecture

We move from observed-variable causal graphs to **latent structures**.

What comes next?

- latent variable models,
- factor models and mixtures,
- latent trees,
- and the bridge from graphical models to neural-network architectures.